# Week 2 - Discovering the Atmosphere 2: Practice Questions

*EART23001 Atmospheric Physics and Weather*

**Workload guide.** The shaded question headings show the intended workload:

- **IN CLASS**: selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY**: expected independent practice after the session.
- **EXTRA CHALLENGE**: optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

## YOU SHOULD TRY Question 1: Reading atmospheric composition quantitatively

Near the surface, dry air is approximately 78.08% $N_2$, 20.95% $O_2$ and 0.93% Ar by mole fraction. Suppose background $CO_2$ is 420 ppm by mole fraction.

1. Convert 420 ppm to a percentage.
2. Out of $10^6$ air molecules, approximately how many are $CO_2$ molecules?
3. Why is water vapour normally excluded when quoting the composition of "dry air"?
4. The abundance of $CO_2$ is far smaller than that of $N_2$ or $O_2$. Give one physical reason why abundance alone does not determine atmospheric importance.

## IN CLASS Question 2: An exponential density profile

Assume an isothermal atmosphere with

$$\rho(z) = \rho_0 \exp\left(-\frac{z}{H}\right),$$

where $\rho_0 = 1.20\ \mathrm{kg\,m^{-3}}$ and $H = 8.0$ km.

1. Calculate the density at 10 km.
2. Calculate the density at 50 km.
3. By what factor has the density fallen between the surface and 50 km?
4. State one reason why this simple expression should not be extrapolated uncritically to several hundred kilometres altitude.

## EXTRA CHALLENGE Question 3: How high before density is only ten per cent?

Using the same exponential model as in Question 2, calculate the altitude at which the density is 10% of its sea-level value. Then calculate the altitude at which it is 1% of its sea-level value.

## IN CLASS Question 4: Diagnose the atmospheric layers from a temperature profile

The table gives a deliberately simplified mean temperature profile.

| Altitude (km) | 0 | 10 | 20 | 50 | 85 |
|---|---|---|---|---|---|
| Temperature (°C) | 15 | -55 | -55 | 0 | -90 |

1. Calculate the mean lapse rate between 0 and 10 km.
2. Identify the broad atmospheric layer represented by 0–10 km.
3. Which layer is represented by the temperature increase from 20 to 50 km?
4. What feature near 10–20 km separates these two layers?
5. Why can ozone absorption produce a temperature increase with height in the stratosphere?

## YOU SHOULD TRY Question 5: Same molecule, different altitude: ozone

Two instruments report enhanced ozone. Site A is a surface monitoring station in a polluted summer air mass. Site B is a balloon instrument at 25 km altitude.

1. At which site is enhanced ozone primarily a health/air-quality concern?
2. At which site does ozone play its major protective role against solar ultraviolet radiation?
3. State the physical reason for each answer in one or two sentences.

## YOU SHOULD TRY Question 6: Why does the Antarctic ozone hole develop in spring?

The Antarctic ozone hole results from a sequence of linked physical and chemical conditions:

CFC-derived chlorine in the stratosphere ⟶ cold, isolated winter vortex and PSCs ⟶ chlorine is chemically primed on PSC surfaces ⟶ spring sunlight activates chlorine ⟶ rapid catalytic ozone loss.

Use this process chain to answer the following questions.

1. Explain in one sentence why the polar vortex is important.
2. Explain in one sentence why PSCs are important.
3. Why does the most rapid ozone loss occur after sunlight returns in spring rather than during the dark Antarctic winter?
4. What eventually limits the severe seasonal ozone loss as spring progresses?

**Modelling connection.**

The following two short problems preview parcel and cloud physics that will be developed in more detail later in EART23001. Students who also take EART22001 will meet related ideas in a parcel-model practical, but no knowledge of that unit is assumed here.

## IN CLASS Question 7: A rising parcel: pressure, cooling and saturation (preview)

A dry air parcel starts at $p_1 = 1000$ hPa and $T_1 = 290$ K and rises to a level where $p_2 = 850$ hPa. Treat the ascent as dry adiabatic and use

$$T_2 = T_1 \left(\frac{p_2}{p_1}\right)^{R_d/c_p}, \qquad \frac{R_d}{c_p} = 0.286.$$

At the initial level the water-vapour partial pressure is $e = 12.0$ hPa. For this preview, assume $e$ changes little before saturation is reached. Use the supplied values $e_s(290\text{ K}) = 19.2$ hPa and $e_s(T_2) = 11.7$ hPa.

1. Calculate $T_2$.
2. Calculate the initial relative humidity, $RH = 100e/e_s$.
3. Using the supplied $e_s(T_2)$, calculate the saturation ratio $\mathcal{S} = e/e_s$ at the upper level and the supersaturation $s = (\mathcal{S} - 1) \times 100\%$.
4. Explain in one or two sentences why rising motion can lead to cloud formation.

## YOU SHOULD TRY Question 8: Same liquid water, different droplet number (preview)

A model cloud contains the same liquid-water content in two simulations. Simulation B has eight times the droplet number concentration of Simulation A. Assume identical spherical droplets within each simulation.

1. Using "number × average droplet mass", explain why constant liquid-water content implies $N_d r^3 = \text{constant}$.
2. By what factor does mean radius $r$ change when $N_d$ increases by a factor of 8?
3. Total projected droplet area is proportional to "number × average projected area", $N_d r^2$. By what factor does the total projected area change?
4. Show that, at fixed liquid-water content, total projected droplet area is proportional to $N_d^{1/3}$. Why is this useful for thinking about the interaction of a cloud with solar radiation?